

Mapping groundwater-dependent ecosystems shows the need for more protection globally

Ecosystems reliant on groundwater are vulnerable to water depletion and climate change, but often their locations are unknown. A map of groundwater-dependent ecosystems across global drylands provides essential information to advance the protection of these biologically diverse ecosystems and the rural societies that depend on them.

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The mission

Many diverse ecosystems rely on groundwater for a stable water supply. Species can rely directly or indirectly on groundwater for some or all their water requirements, depending on a multitude of factors such as their life-history stage and the local climate or hydrology. These groundwater-dependent ecosystems span terrestrial and aquatic realms, including mountain meadows, rivers, wetlands, estuaries, terrestrial forests, marine springs and even ecosystems in aquifers¹. They are often biological hotspots, supporting rare and endemic species and providing crucial refuges for migratory and endemic species during droughts².

Many of these ecosystems are increasingly at risk owing to global groundwater depletion and a lack of awareness of their importance. What's more, where exactly they are is often unknown. Mapping their locations is thus an important first step towards better protecting them.

The discovery

Most ecosystem-mapping efforts involve time-consuming local data collection and field studies. To map groundwater-dependent ecosystems across global drylands, we used publicly available satellite imagery and machine learning to produce a map with a spatial resolution of around 30 metres.

In the absence of a comprehensive global database of groundwater levels, we used an inference-based approach, which assumed that ecosystems with access to groundwater would appear as blue or green 'islands' during the dry season. We selected globally available satellite data on vegetation greenness, leaf water content, presence of open water bodies, evapotranspiration rates relative to precipitation and land-surface temperatures. This approach enabled us to conservatively map ecosystems likely to be reliant on groundwater, create a reproducible methodology for periodic or refined mapping efforts and identify key locations that should be prioritized.

Our mapping reveals that there are probably 8.3 million square kilometres of groundwater-dependent ecosystems in global drylands. They are particularly contiguous and extensive across pastoral landscapes, such as those in Central Asia and the Sahel in Africa, which have lower rates of groundwater depletion.

Regions dominated by agricultural land and that have a history of groundwater pumping, such as Australia and North America, have, by contrast,

more-fragmented ecosystems, presumably because of ecological degradation and losses in the decades since the pumping commenced. Nevertheless, more than half of the groundwater-dependent ecosystems we mapped are in regions with declining groundwater storage, raising concerns about long-term ecosystem viability. Only one-fifth of ecosystems have some degree of legal protection (Fig. 1), although many of those protection measures, such as conservation efforts or sustainable groundwater legal frameworks, are ineffective^{3,4}.

Future directions

Global initiatives to combat climate change, biodiversity loss and humanitarian crises often overlook the significance of groundwater in supporting species, habitats and rural peoples' livelihoods. Groundwater protections for ecosystems are explicit in only a handful of sustainable water policies and there are still notable policy gaps globally⁵. Even laws that limit groundwater development or call for sustainable groundwater-management planning might be implemented ineffectively, or be unworkable in politically unstable regions. In these situations, it might be necessary to implement other local, national or international policies or humanitarian efforts.

Working at the global scale with remote sensing and machine-learning approaches has its limitations. Our map is intended to indicate where groundwater-dependent ecosystems are likely to exist across global drylands, and to provide a starting point for more-refined mapping and verification efforts using local data. The random-forest machine-learning approach we used is statistical rather than deterministic, so our map is accompanied by a probability layer that indicates the likelihood, between 0 and 100%, that each pixel is a groundwater-dependent ecosystem. Therefore, the map should be used with care, because the absence of groundwater-dependent ecosystems on our map does not preclude the possibility of them existing.

Even so, the map provides crucial information for subnational, national and intergovernmental organizations to prioritize, conceptualize and develop policy and protection mechanisms to safeguard these important ecosystems and the communities that depend on them.

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EXPERT OPINION



This interesting study does much to highlight the significance and spatial extent of surface groundwater-dependent ecosystems globally. It provides, by way of a case study, a valuable narrative of the role of governance in the management of these ecosystems. The global

analysis is novel and provides a wider context for the numerous regional-scale mapping programmes that have been done previously.”

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FIGURE

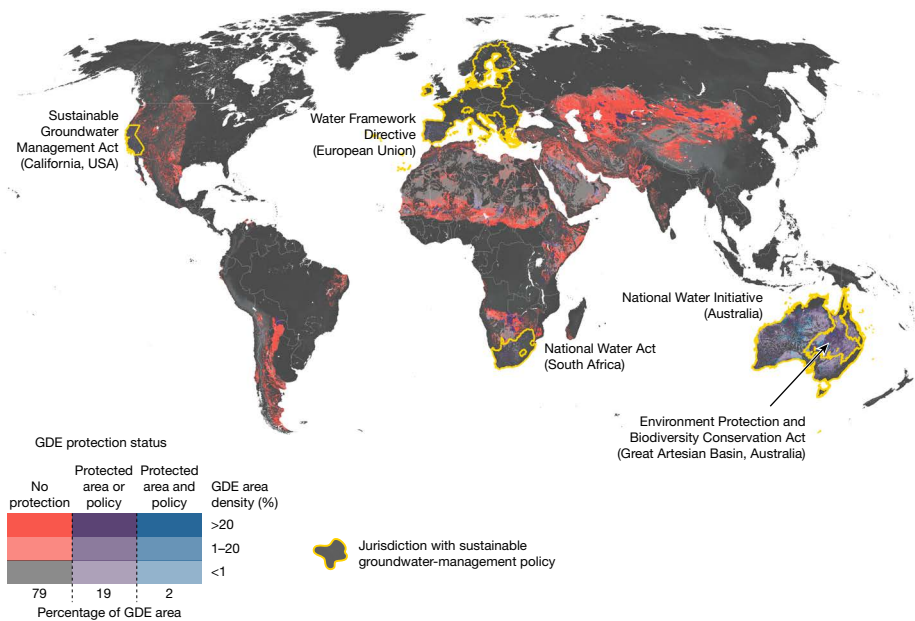


Figure 1 | Groundwater-dependent dryland ecosystems lack protection. Globally, 79% of mapped groundwater-dependent ecosystems have no legal protection (red), with the remaining 21% having some degree of protection (blue and purple). Mapped groundwater-dependent ecosystems (GDE) with high protection status include World Database on Protected Areas (WDPA) categories Ia (strict nature reserve), Ib (wilderness area), II (national park) and III (national monument or feature). Mapped ecosystems with low protection status (light blue) include WDPA categories IV (habitat or species management area), V (protected landscape or seascape) and VI (protected area with sustainable use of natural resources) and jurisdictions with sustainable groundwater-management policies that include groundwater-dependent ecosystems. Credit: base map, © USGS HydroSHEDS (2024). Nature publications remain neutral with regard to contested jurisdictional claims in published maps. Rohde, M. M. *et al.*/*Nature* (CC BY 4.0).

BEHIND THE PAPER

This research project started when I was chair of the global groundwater group in the non-profit organization The Nature Conservancy, working to advance the science, policy and practice of groundwater conservation. Many parts of the organization were lacking maps of groundwater-dependent ecosystems, which would have helped us to understand potential groundwater impacts on ecosystems, advocate protective policies and implement conservation projects.

My suggestion was to map them using remote-sensing data — which turned into a large undertaking, involving an international and highly interdisciplinary team of more than 20 people. Our project was truly collaborative and a labour of love. Without the support of every member of the team none of this work would have been possible.

M.M.R.

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FROM THE EDITOR

This study fills an important knowledge gap about where groundwater-dependent ecosystems can be found in the world. With increasing pressure on freshwater resources from global warming and changing land-use practices, these high-resolution maps provide valuable data for formulating robust groundwater-protection policies in the future.

Editorial team, Nature